Elliptic Curve Cryptography

Gemma Powell

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1 Introduction

Imagine we wish to communicate some highly private and sensitive information. How can it be ensured that the contents will not be discovered by an intruder? Although absolute security can never be guaranteed, we can decrease the likelihood an eavesdropper decrypts the message. Cryptography is the solution.

In a population reliant on digital communication, cryptography is the cornerstone in providing confidentiality and protection, encrypting messages using algorithms with keys known only to the senders and recipients. Not only does it ensure any information is understood by only the intended parties, cryptography also allows for verification via digital signatures. This means a recipient can comfortably assume a message is from the correct sender and has not been tampered with [2, p. 5]. Cryptographic methods are as old as civilisation itself, but elliptic curve cryptography (ECC) has been developed more recently. Introduced by Neal Koblitz [7] and Victor Miller [9] in the 1980s, elliptic curve cryptography uses points on a curve in order to encrypt and decrypt data, and is now the leading choice for many modern cryptosystems. This essay will cover the fundamentals behind elliptic curve cryptography, introduce various cryptographic protocols, and evaluate the effectiveness of elliptic curves in providing secure encryption.

2 Elliptic curves

2.1 Finite fields

Within cryptography, only elliptic curves defined over finite fields are used to ensure predictable behaviour of points on the curve. These fields commonly fall into two categories, both denoted by $\mathbb{F}_q$. The first occurs when $q = p$, where p is a large prime number, and the second when $q = 2^m$, where m is a large integer [10, p. 230].

For a large prime p, $\mathbb{F}_p$ is called a prime field. This will be the set of integers modulo p. For example, if $p = 3$, $\mathbb{F}_3 = \{0, 1, 2\}$.

For $q = 2^m$, $\mathbb{F}_{2^m}$ is called a binary field. This can be described as an m-dimensional vector space over $\mathbb{F}_2$ [10, pp. 234–236]. For any chosen basis of the form $\{\alpha_0, \ldots, \alpha_{m-1}\}$, each element in $\mathbb{F}_{2^m}$ can be written as a linear combination of the basis elements. There are many different possible bases, but the two most frequently used are polynomial and normal bases.

A polynomial basis consists of powers of x up to $m - 1$, meaning the basis is of the form $\{x^{m-1}, x^{m-2}, \ldots, x^2, x, 1\}$. Each element $a \in \mathbb{F}_{2^m}$ can be written as

$$a = a_{m-1}x^{m-1} + \cdots + a_1 x + a_0,$$

where $a_i \in \mathbb{F}_2$ for every $i \in \{0, 1, \ldots, m - 1\}$. This element a therefore has binary representation $(a_{m-1}a_{m-2} \cdots a_0)$ in $\mathbb{F}_{2^m}$. Thus, within the vector

space, the multiplicative identity is given by $(0\ldots01)$ and the additive identity is $(0\ldots0)$. We can therefore represent $\mathbb{F}_{2^m}$ as $\mathbb{F}_2[X]/f\mathbb{F}_2[X]$, where f is an irreducible polynomial of degree m in the polynomial ring $\mathbb{F}_2[X]$. This f is known as the reduction polynomial for the field $\mathbb{F}_{2^m}$ [4, pp. 417–418].

A normal basis is of the form $\{\beta, \beta^2, \ldots, \beta^{2^{m-1}}\}$, meaning each $a \in \mathbb{F}_{2^m}$ can be written as

$$a = \sum_{i=0}^{m-1} a_i \beta^{2^i},$$

where $a_i \in \mathbb{F}_2$ for each $i \in \{0, 1, \ldots, m-1\}$. The binary representation of a is now $(a_0 a_1 \cdots a_{m-1})$. In this case, the multiplicative identity is $(1 \cdots 1)$ and the additive identity is $(0 \cdots 0)$.

The structure of a finite field ensures any calculations necessary are both predictable and reliable. Working with only a finite number of points on any given curve is much more manageable, which is essential when implementing a cryptographic system.

2.2 Elliptic curves over finite fields

Before introducing elliptic curves, we must first introduce an affine plane. If K is a field, then the affine plane over K is denoted by $\mathbb{A}^2(K)$, where

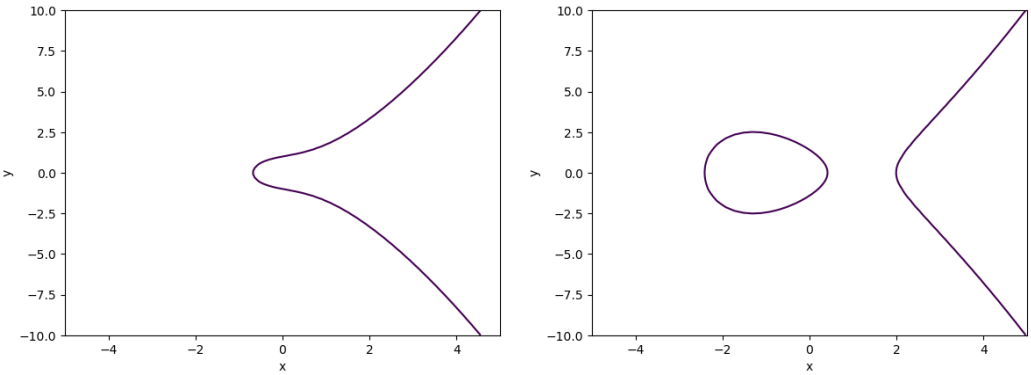
$$\mathbb{A}^2(K) = \{(x, y) \mid x, y \in K\}.$$

An elliptic curve E over a finite field is a subset of the affine plane over that field [4, p. 438].

Definition 2.1 (Weierstrass equation)**.** [4, p. 446] A Weierstrass equation is an equation of the form

$$y^2 + a_1xy + a_2y = x^3 + a_3x^2 + a_4x + a_5,$$

where a_1, a_2, a_3, a_4, a_5 are constants. An elliptic curve E over a finite field K is a subset of $\mathbb{A}^2(K)$ defined by a Weierstrass equation, with coefficients in K. The coefficients are commonly accompanied by some conditions.



(a) $y^2 = x^3 + x + 1$ (b) $y^2 = x^3 - 5x + 2$

Figure 1: Illustrations of elliptic curves over $\mathbb{R}$.

Within any elliptic curve, $\mathcal{O}$ is defined as the point lying on every vertical line, or the point at infinity. This point is crucial in elliptic curve cryptography, playing a fundamental role in all calculations.

Definition 2.2 (Elliptic curve over a prime field)**.** [4, p. 90] An elliptic curve over a finite field $\mathbb{F}_p$, where p is a large prime, is denoted by $E(\mathbb{F}_p)$, where

$$E(\mathbb{F}_p) = \{(x, y) \in \mathbb{F}_p^2 \mid y^2 \equiv x^3 + ax + b \pmod{p}\} \cup \{\mathcal{O}\}.$$

And, $a, b \in \mathbb{F}_p$ satisfy $4a^3 + 27b^2 \not\equiv 0 \pmod{p}$.

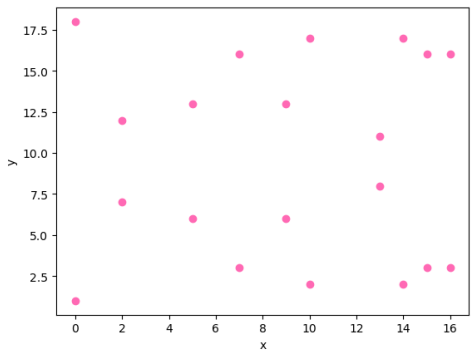


Figure 2: The elliptic curve $y^2 = x^3 + x + 1$ over the prime field $\mathbb{F}_{19}$

Definition 2.3 (Elliptic curve over a binary field)**.** [4, p. 90] An elliptic curve over a finite field $\mathbb{F}_{2^m}$, where m is a large integer, is denoted by $E(\mathbb{F}_{2^m})$, where

$$E(\mathbb{F}_{2^m}) = \{(x, y) \in \mathbb{F}_{2^m}^2 \mid y^2 + xy = x^3 + ax^2 + b\} \cup \{\mathcal{O}\}.$$

And, $a, b \in \mathbb{F}_{2^m}$ satisfy $b \neq 0$.

The conditions on these coefficients ensure the curve does not intersect itself at any point, meaning computations needed in cryptography are plausible. Elliptic curves which appear similar to those in figure 1 are usually what we would imagine, but within cryptography we actually study those like figure 2, where a curve is comprised of individual points.

2.3 Point addition on an elliptic curve

Within the set of points on an elliptic curve, we can define a unique addition. This addition forms the binary operation to create an abelian group. Let $E(\mathbb{F}_q)$ be the set of points on the elliptic curve E defined over $\mathbb{F}_q$. Firstly, for a point $P = (x_P, y_P) \in E(\mathbb{F}_q)$ we define $-P$ as the reflection of P in the x-axis. Therefore, if $q = p$ is a large prime then $-P = (x_P, -y_P)$, and if $q = 2^m$ for an integer m then $-P = (x_P, x_P + y_P)$.

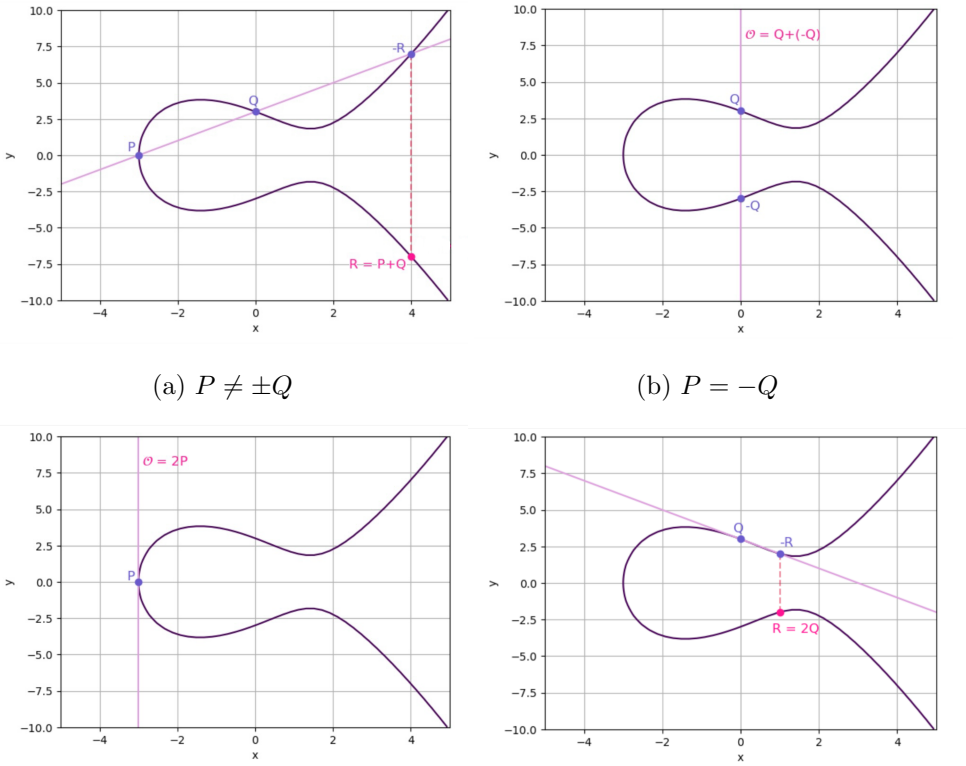
For $P, Q \in E(\mathbb{F}_q) \setminus \mathcal{O}$, we will define $P + Q$ geometrically. Silverman introduces the concept of this in his book [11, pp. 280–282]. Given these two points we must construct a line through both, however this will depend on the specific values of P and Q.

When $P \neq \pm Q$, the line through P and Q will intersect the curve E at one other point, which we denote by $-R$. In this case, we reflect $-R$ in the x-axis to obtain $P + Q = R$.

When $P = -Q$, the line will not intersect the curve at any other point. In fact, the line is vertical, and we define $P + Q = \mathcal{O}$, the point at infinity. In this case, we can also write $P + Q = P + (-P) = P - P$.

When $P = Q$, the line through both points is simply the tangent to the elliptic curve at P. If this tangent is parallel to the y-axis, $2P = P + P = \mathcal{O}$. If it is not parallel to the y-axis, it will intersect E at another point. Letting this point be $-R$, we have $2P = R$.

If we let $Q = \mathcal{O}$, then we define $P + \mathcal{O} = \mathcal{O} + P = P$ for any $P \in E(\mathbb{F}_q)$. Thus, $\mathcal{O}$ will act as the identity element for point addition.



(a) $P \neq \pm Q$

(b) $P = -Q$

(c) $P = Q$: case 1

(d) $P = Q$: case 2

Figure 3: The graphical representation of point addition of two points P and Q on the elliptic curve $y^2 = x^3 - 6x + 9$.

In figure 3, we observe graphically the point addition of P and Q for each case discussed.

2.4 The algebra of point addition in $\mathbb{F}_p$

When working with the field $\mathbb{F}_p$ for a large prime p, the calculations involved in point addition are described by Omondi in his book [10, pp. 228–229]. If $P = (x_P, y_P)$ and $Q = (x_Q, y_Q)$ are points on an elliptic curve E over $\mathbb{F}_p$, then for $P \neq \pm Q$ the gradient of the line passing through both P and Q is given by

$$\lambda = \frac{y_Q - y_P}{x_Q - x_P}. \tag{2.1}$$

In the event $(y_Q - y_P)$ and $(x_Q - x_P)$ are coprime and $(x_Q - x_P) \neq 1$, λ is not an integer. We define $\lambda = (y_Q - y_P) \cdot (x_Q - x_P)^{-1}$ where $(x_Q - x_P)^{-1}$ is the inverse of $(x_Q - x_P)$ modulo p. Then, the equation of the line through

both P and Q is given by

$$y = y_P + \lambda(x - x_P). \tag{2.2}$$

Recalling that E is of the form $y^2 = x^3 + ax + b$, we can find the intersection $R = (x_R, y_R)$ by setting the two equations equal to one another. We see that

$$(y_P + \lambda(x - x_P))^2 = x^3 + ax + b. \tag{2.3}$$

Upon expansion of the left hand side of (2.3), we get

$$x^3 + x^2(-\lambda^2) + x(a + 2\lambda^2 x_P - 2\lambda y_P) + b + 2\lambda y_P x_P - \lambda^2 x_P - y_P^2 = 0.$$

This is a cubic equation in x, with roots x_P, x_Q, and x_R. Recalling that for cubic equations the sum of the roots is equal to the negative of the coefficient of the x^2 term, it is clear that

$$x_R = \lambda^2 - x_P - x_Q. \tag{2.4}$$

Then, (2.2) tells us that

$$y_R = \lambda(x_P - x_R) - y_P. \tag{2.5}$$

Thus for $P, Q \in \mathbb{F}_p$, $P + Q = (\lambda^2 - x_P - x_Q, \lambda(x_P - x_R) - y_P)$.

In the case $P = Q$, we must find the tangent of the line passing through P. Using implicit differentiation on the equation of E,

$$2y\frac{\mathrm{d}y}{\mathrm{d}x} = 3x^2 + a,$$

which implies

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{3x^2 + a}{2y}.$$

So the gradient of the line through P is given by

$$\lambda = \frac{3x_P^2 + a}{2y_P} \tag{2.6}$$

Then, using (2.4) and (2.5), we see that

$$x_R = \lambda^2 - 2x_P, \tag{2.7}$$

$$y_R = \lambda(x_P - x_R) - y_P. \tag{2.8}$$

This means $2P = R = (x_R, y_R) = (\lambda^2 - 2x_P, \lambda(x_P - x_R) - y_P)$. Formulae for point addition on elliptic curves over $\mathbb{F}_{2^m}$ can be derived similarly, but will not be covered here since the calculations are tedious.

Example 2.4 (Point addition)**.** We begin with a simple elliptic curve given by

$$E_1 \colon y^2 = x^3 + 6x + 9$$

over the prime field $\mathbb{F}_{29}$. Since $3^2 = 9 \equiv 0^3 + 6 \cdot 0 + 9 \pmod{29}$ and $4^2 = 16 \equiv 1^3 + 6 \cdot 1 + 9 \pmod{29}$, two points on this curve are $(0, 3)$ and $(1, 4)$. Using (2.1), the gradient of the line through these points is $\lambda = \frac{4-3}{1-0} = 1$. Now, with (2.4) and (2.5) we can obtain $(0, 3) + (1, 4) = (0, -3)$ which is equivalent to $(0, 26)$ modulo 29.

Example 2.5 (Point doubling)**.** Now consider the elliptic curve

$$E_2 \colon y^2 = x^3 + 2x + 13$$

over the prime field $\mathbb{F}_{17}$. As $4^2 = 16 \equiv 1^3 + 2 \cdot 1 + 13 \pmod{17}$, a point on this curve is $(1, 4)$. Letting $P = (1, 4)$, we will compute $2P$. Using (2.6), the gradient of the tangent at P is given by $\lambda = \frac{3 \cdot 1 + 12}{2 \cdot 4} = \frac{5}{8}$. The inverse of 8 modulo 17 is 15, so we get $\lambda = 5 \cdot 15 = 75 \equiv 7 \pmod{17}$. Assuming $2P = (x_R, y_R)$, by (2.7) we have $x_R = 47 \equiv 13 \pmod{17}$ and by (2.8) we have $y_R = -88 \equiv 14 \pmod{17}$. Therefore, $2(1, 4) = (13, 14)$.

2.5 The elliptic curve group

To encrypt and decrypt messages in an elliptic curve cryptosystem, we require calculations within a specific elliptic curve group with binary operation given by point addition.

Theorem 2.6. [4, p. 450] *Given an elliptic curve E over a finite field $\mathbb{F}_q$, we have that $(E(\mathbb{F}_q), +)$ is an abelian group. The identity element is $\mathcal{O}$ and the inverse of a point $P \in E(\mathbb{F}_q)$ is given by $-P$,the reflection of P in the x-axis.*

Proof. We must show that the four defining properties of a group hold. From the definition of point addition on a curve, we know that for any two points $P, Q \in E(\mathbb{F}_q)$, $P + Q$ is another point on the curve. This means $P + Q \in E(\mathbb{F}_q)$, and $E(\mathbb{F}_q)$ is closed under $+$.

The full proof of associativity involves tedious algebraic manipulation, so will not be covered here. We instead give a sketch. First, we express point addition algebraically, as in Section 2.4. Then, to prove $(P + Q) + R = P + (Q + R)$ for all $P, Q, R \in E(\mathbb{F}_q)$, we use (2.4), (2.5), (2.7) and (2.8) recursively to achieve the result.

We know that for any $P \in E(\mathbb{F}_q)$, $P + \mathcal{O} = \mathcal{O} + P = P$, so the identity element is given by $\mathcal{O}$, which lies on every elliptic curve.

Finally, point addition is defined such that $P + (-P) = (-P) + P = \mathcal{O}$, so the inverse of a point $P \in E(\mathbb{F}_q)$ is given by $-P$, the reflection of P in the x-axis. Since elliptic curves are symmetric about the x-axis, we must have $-P \in E(\mathbb{F}_q)$. Thus, $(E(\mathbb{F}_q), +)$ forms a group.

Since any line passing through two points P and Q is also the line passing through Q and P, we can deduce that point addition is commutative. Thus, the group is abelian. $\square$

The elliptic curve group provides a foundation on which we can build many secure cryptographic protocols. Elliptic curve cryptography typically involves point addition within large cyclic subgroups of this abelian group [4, p. 91]. To establish which generator to use for this cyclic subgroup, it is essential to know the order of an element on an elliptic curve. For a system to be as efficient as possible, we require a point with a large prime order.

Definition 2.7. [13, p. 337] The order of a point $P \in E(\mathbb{F}_q)$ is the smallest positive integer n such that $nP = \mathcal{O}$ in the sense of point addition n times.

Example 2.8 (The order of an element)**.** Consider the elliptic curve over the finite field $\mathbb{F}_{11}$ given by

$$E\colon y^2 = x^3 + x - 1.$$

We wish to compute the order of $(1, 1)$ on the curve. Firstly,

$$2(1, 1) = (2, -3),$$

then,

$$3(1, 1) = (1, 1) + (2, -3) = (13, 47) \equiv (2, 3) \pmod{11}.$$

And,

$$4(1, 1) = (1, 1) + (2, 3) = (1, -1).$$

Since $(1, -1) = -(1, 1)$, we have

$$5(1, 1) = (1, 1) + (1, -1) = \mathcal{O}.$$

Thus, the order of $(1, 1)$ on E over $\mathbb{F}_{11}$ is 5.

3 Elliptic curve cryptography

Using repeated point addition on elliptic curves, it is possible for two parties to communicate a message which can only be deciphered by those intended, a method which forms the basis of elliptic curve cryptography.

3.1 The elliptic curve discrete logarithm problem

We first recall the discrete logarithm problem (DLP) from MA132 Foundations, which is used widely within cryptography.

Problem 3.1 (The discrete logarithm problem)**.** [11, p. 63] *Given a large prime p and non-zero integers a and b, we wish to find an x such that $a^x \equiv b \pmod{p}$.*

The DLP ensures that once an x is chosen privately and $a^x \pmod p$ is calculated, it is computationally infeasible for an intruder to find this x even when both a and b are known. The elliptic curve discrete logarithm problem (ECDLP) is the elliptic curve alternative to the DLP.

Problem 3.2 (The elliptic curve discrete logarithm problem)**.** [4, p. 91] *We are given a base point G of order n on an elliptic curve E over a finite field $\mathbb{F}_q$. Looking at the cyclic group generated by G, $\langle G \rangle$, and a point $P \in \langle G \rangle$, we want to determine an integer k such that $0 \leq k \leq n-1$ and $P = kG$. We denote this k by $k = \log_G(P)$.*

The ECDLP is one of the most difficult problems to solve in cryptography, ensuring systems which rely on it are highly secure [6].

Theorem 3.3. [13, p. 316] *For a prime number p and an elliptic curve E over the field $\mathbb{F}_p$, the fastest known algorithm to solve the elliptic curve discrete logarithm problem takes approximately $\sqrt{p}$ steps.*

Since point addition involves non-linear calculations which can be challenging to reverse, it is considered infeasible to compute a solution to the ECDLP over large prime fields. But, the calculation of kG for a given base point G of order n and an integer $k \in [0, n-1]$ is fairly simple, making elliptic curve cryptography an ideal candidate for protecting sensitive information.

3.2 Hash functions

From digital signatures to authentication, hash functions have various uses in cryptography. A hash function takes an arbitrary-length input and produces a fixed-length output to then be used in calculations, which can only be carried out on specific input lengths. For example, it could convert a file, video or message, all of varying lengths, into a fixed 256-bit length ready for download.

Hash functions must be comprised of three strong foundations, as described by Wong in his book [14, p.48]. The first is pre-image resistance, which guarantees the hash function cannot be reversed to reveal the initial message. We then require second pre-image resistance, ensuring that given an input and its output, we cannot find another input which maps to the same output under the hash function. Finally, a hash function requires collision resistance. This ensures that an intruder is unable to find two different inputs that map to the same output. This differs to second pre-image resistance in that it does not depend on a specific output and input. These properties ensure that even a slight modification in a message results in a completely different hash output. If messages did not have unique hash outputs, an intruder could send a fraudulent message that would be interpreted by a recipient as being authentic, misleading them. A sender may accompany their message with a hash value, allowing any recipient to verify the communication has not been tampered with.

3.3 Message-embedding functions

Message-embedding functions are a way of mapping messages to be encrypted onto points on the selected elliptic curve. There are various methods that can be used to encode the message $m \in \{0, 1, \ldots, q-1\}$, the one we will cover here is a probabilistic function which is described by Delfs and Knebl in their book [4]. We begin with an elliptic curve E over a finite field $\mathbb{F}_q$.

Protocol 3.4 (Probabilistic message-embedding function)**.** [4, p. 101]

(1) Select a random $r \in \mathbb{F}_q^*$, the set of units in $\mathbb{F}_q$.

(2) Find an integer x such that $x \equiv rm \pmod{q}$.

(3) If there is a y such that $(x, y) \in \langle G \rangle$, the cyclic subgroup generated by the base point G, then the message m is mapped to the point (x, y). Otherwise, repeat from step (1) with a new r.

The probability we find a sufficient y in step (3) of Protocol 3.4 is $n/2\#E(\mathbb{F}_q)$, which implies that, when working over a large finite field with many points on the curve, this process may take a significant amount of time. Thus, this may not be the most efficient message-embedding function. Using this procedure, any message m can be mapped to a point on the curve in order to be encrypted in a specific cryptosystem.

3.4 Elliptic curve cryptosystems

There are numerous ways to encrypt and decrypt a message m using points on an elliptic curve. In any cryptosystem, Alice and Bob are two entities wishing to perform some form of communication, and Eve is an eavesdropper hoping to spy on or tamper with this communication. Alice and Bob will decide on certain public information, commonly this consists of a specific elliptic curve E over a finite field $\mathbb{F}_q$ and a base point G of order n in $E(\mathbb{F}_q)$. These are known as the domain parameters, and they will be public so any system implemented must ensure that even with access to the information, Eve cannot discover the message.

3.4.1 Digital signatures

A digital signature is a method of 'signing' a message so that it can be verified by any recipient. It must not have the possibility of being forged by an impostor; altered in any way; or associated with a different message [10, p. 249]. To begin, Alice and Bob agree on the common domain parameters. They also both compute $e = h(m)$, where h is a secure hash function and m is a message. Alice wishes to sign this message so that any recipient can confirm her identity as the sender.

Protocol 3.5 (Producing a signature)**.** [10, pp. 249–251]

(1) Alice randomly selects integers k and d such that $1 \leq k, d \leq n-1$.

(2) Alice computes $(x_P, y_P) = P = kG$ and $H = dG$.

(3) Alice computes $r = x_P \pmod{n}$.

(4) If $r = 0$, then repeat from (1).
Otherwise, Alice can compute $s = k^{-1}(e + rd) \pmod{n}$, where k^{-1} is the inverse of k modulo n.

(5) If $s = 0$, repeat from (1).
Otherwise, the digital signature is given by (r, s).

Alice then sends the message m to Bob, along with the digital signature (r, s). To verify any message Alice has signed, Bob begins by computing

$$u = s^{-1}e \pmod{n} \quad \text{and} \quad v = s^{-1}r \pmod{n}.$$

Then, he can compute $(x_Q, y_Q) = Q = uG + vH$. The signature is valid if $r \equiv x_Q \pmod{n}$.

Example 3.6 (Elliptic curve digital signature)**.** Consider the elliptic curve over $\mathbb{F}_5$ given by

$$E\colon y^2 = x^3 + x + 1.$$

Let $G = (0, 1)$, which has order 7 on E. Alice wishes to send a message m to Bob with a digital signature he can verify, to keep calculations simple let $e = 2$.

(1) Alice chooses $k = 4$ and $d = 3$.

(2) Then, $P = 4(0, 1) = (3, 1)$ and $H = 3(0, 1) = (2, 4)$.

(3) Now, $r = 3 \pmod{7} = 3$.

(4) Since $4 \cdot 2 = 8 \equiv 1 \pmod{7}$, $k^{-1} = 2$.
Thus, $s = 2 \cdot (2 + 3 \cdot 3) \pmod{7} = 22 \pmod{7} = 1$.

(5) The digital signature is therefore given by $(3, 1)$.

Then, Alice can send the message m to Bob along with the signature $(3, 1)$ and Bob can verify as follows. Clearly $s^{-1} = 1$, so $u = 1 \cdot 2 \pmod{7} = 2$ and $v = 1 \cdot 3 \pmod{7} = 3$. Bob can now calculate $Q = uG + vH$ as $2(0, 1) + 3(2, 4) = (3, 1)$. The message can be verified as $3 \equiv 3 \pmod{7}$, meaning Bob can safely assume Alice was the sender.

3.4.2 Elliptic curve Diffie-Hellman key exchange

The Diffie-Hellman key exchange was first published in 1976 by Whitfield Diffie and Martin Hellman [5]. In the late 1980s, the method was then adapted to utilise elliptic curves, creating the elliptic curve Diffie-Hellman key exchange. For a key exchange, Alice and Bob agree, via any communication channel, on a 'secret' to either be used as a key or to generate a key. However, any communication can be eavesdropped so they cannot simply send it to one another. Therefore, they must both independently generate the key in a way that any eavesdropper cannot reproduce. To achieve this, each party needs both a private and public key [14, p. 100]. Before any communication, Alice and Bob must first agree on the usual domain parameters.

Protocol 3.7 (Elliptic curve Diffie-Hellman key exchange)**.** [10, p. 247]

(1) Alice randomly selects an integer n_A such that $1 \leq n_A \leq n-1$, and computes $P_A = n_A G$. Alice's public key is P_A and her private key is n_A.

(2) Bob randomly selects an integer n_B such that $1 \leq n_B \leq n-1$, and computes $P_B = n_B G$. Bob's public key is P_B and his private key is n_B.

(3) Alice sends P_A to Bob, and Bob sends P_B to Alice.

(4) Alice computes $n_A P_B$ and Bob computes $n_B P_A$, which both correspond to the secret key $n_A n_B G$.

The secret key $n_A n_B G$ in Protocol 3.7 can only be obtained with either Alice or Bob's private key, each of which is known only to the respective party. If Eve were to find both P_A and G, in order to retrieve n_A she would need to solve the ECDLP which we recall is deemed infeasible to calculate. Thus, as long as the base point G has order large enough, the elliptic curve Diffie-Hellman key exchange provides secure transmission of information that is very difficult to successfully eavesdrop.

Example 3.8 (Elliptic curve Diffie-Hellman Key exchange)**.** Let E be the curve $y^2 = x^3 + 2x + 13$ over $\mathbb{F}_{17}$, and $G = (1, 4)$.

(1) Alice chooses $n_A = 2$ and computes $P_A = 2(1, 4) = (13, 14)$. She then sends P_A to Bob.

(2) Bob chooses $n_B = 5$ and computes $P_B = 5(1, 4) = (15, 1)$. He then sends P_B to Alice.

(3) Both Alice and Bob can compute the shared key as $2P_B = 5P_A = (2, 5)$.

3.4.3 Matsumoto-Takashi-Imai system

Similar to the Diffie-Hellman system, the Matsumoto-Takashi-Imai system is a key exchange method in which both parties wish to calculate a shared key. As usual, the domain parameters are agreed upon by Alice and Bob.

Protocol 3.9 (Matsumoto-Takashi-Imai system)**.** [10, p.248]

(1) Alice chooses an integer a such that $1 \leq a \leq n-1$ and computes $P_A = aG$.

(2) Bob chooses an integer b such that $1 \leq b \leq n-1$ and computes $P_B = bG$.

(3) Alice sends P_A to Bob, and Bob sends P_B to Alice.

(4) Alice chooses another integer c such that $1 \leq c \leq n-1$ and computes $Q_A = cP_B$.

(5) Bob chooses another integer d such that $1 \leq d \leq n-1$ and computes $Q_B = dP_A$.

(6) Alice sends Q_A to Bob, and Bob sends Q_B to Alice.

(7) The shared key is given by dcG modulo n. To find this, Alice calculates $a^{-1}cQ_B$ and Bob calculates $b^{-1}dQ_A$. Here, a^{-1} and b^{-1} are the respective inverses of a and b modulo n.

Although it may not seem obvious the computations in step (7) of Protocol 3.9 are equivalent to the shared key, simple calculations can show that they are in fact the same. In Alice's case, we see that since multiplication is commutative modulo n,

$$a^{-1}cQ_B = a^{-1}cdP_A = a^{-1}cdaG = dcG.$$

Likewise for Bob,

$$b^{-1}dQ_A = b^{-1}dcP_B = b^{-1}dcbG = dcG.$$

Compared to the elliptic curve Diffie-Hellman protocol, the Matsumoto-Takashi-Imai system introduces an additional private and public key for both Alice and Bob. This results in the calculation of the shared key involving another step with inverses. While this may improve control and security of the shared secret, it also requires more complex computations which could result in mistakes. In contrast, the Diffie-Hellman approach is much simpler, so is more widely used within cryptography.

3.4.4 The El-Gamal system

Unlike a key exchange, the El-Gamal system does not rely on both parties being in possession of a shared key in order to decrypt a message. Instead, it is a form of message encryption, meaning Alice wishes to encrypt a message which can only be deciphered by Bob [10, p. 244]. To begin, Alice and Bob agree on the common domain parameters. But, they must also decide on a suitable message-embedding function F. Let $\langle G \rangle$ be the cyclic subgroup of $E(\mathbb{F}_q)$ generated by the base point G, all calculations in this cryptosystem will be carried out within $\langle G \rangle$. Alice wishes to encrypt a message m for Bob to decrypt.

Protocol 3.10 (The El-Gamal system)**.** [4, p. 100]

(1) Alice computes $P_m = F(m)$, a point to represent the message m on the curve.

(2) Bob chooses his private key s_B such that $1 \leq s_B \leq n-1$ and calculates his public key $P_B = s_B G$. As it is a public key, P_B is known by Alice.

(3) Alice randomly chooses an integer k such that $1 \leq k \leq n-1$.

(4) Alice computes kG and kP_B.

(5) Alice computes $C_1 = kG$ and $C_2 = P_m + kP_B$.

(6) Alice sends the ciphertext (C_1, C_2) to Bob.

(7) In order to decrypt the ciphertext, Bob computes $s_B C_1 = k s_B G = kP_B$.

(8) Then, Bob computes $C_2 - s_B C_1 = P_m$.

(9) To obtain the plaintext, Bob simply calculates $F^{-1}(P_m) = m$.

Using Protocol 3.10, Alice can send a message to Bob which only he can recover. The decryption process works as point addition is both commutative and associative, ensuring the message will be obtained regardless of the order of operations chosen. Without knowing the value of Bob's private key, it is almost infeasible for an eavesdropper to discover the message m, making the El-Gamal system very secure.

3.5 Weil pairings as a threat to cryptosystems

Silverman et al. described the threat of Weil pairings in their book [13]. They are an example of a bilinear pairing, taking two inputs and giving one output. In this instance, we require inputs in the set of points of a specific order on an elliptic curve. With the outputs we can then solve the ECDLP, which serves as the primary security feature of most cryptosystems.

Definition 3.11. [13, p. 337] Let n be a positive integer, then we denote the set of points of order n on an elliptic curve E by $E[n]$. In other words,

$$E[n] = \{P \in E \mid nP = \mathcal{O}\}.$$

Definition 3.12. [13, p. 338] A function f has a zero α of multiplicity n if we can write

$$f(x) = (x - \alpha)^n g(x)$$

for some function g. A function f has a pole β of multiplicity m if we can write

$$f(x) = \frac{1}{(x - \beta)^m} h(x)$$

for some function h.

To compute the Weil pairing of two points on an elliptic curve, we must first introduce two functions: div and ord. The order of a function f at a point Q is represented by $\mathrm{ord}_Q(f)$, and corresponds to the multiplicity of that point as either a pole or zero of f. If Q is a zero of multiplicity n of f, then $f(Q) = 0$ and $\mathrm{ord}_Q(f) = n > 0$. If Q is a pole of f of multiplicity m, then $f(Q)$ is undefined and $\mathrm{ord}_Q(f) = -m < 0$. Otherwise, if f is both defined and non-zero at Q, we let $\mathrm{ord}_Q(f) = 0$. The divisor function, $\mathrm{div}(f)$ is given by the formal sum

$$\mathrm{div}(f) = \sum_{Q \in E} \mathrm{ord}_Q(f)[Q],$$

where $\mathrm{ord}_Q(f)$ is non-zero for a finite number of points $Q \in E$. Thus, $\mathrm{div}(f)$ depicts the zeros and poles of f along with the relative multiplicities. With these functions we can now calculate the Weil pairing.

Definition 3.13. [13, p. 340] For an elliptic curve E over a finite field $\mathbb{F}_q$, let $P, Q \in E[n]$ and f_P, f_Q be rational functions which satisfy

$$\mathrm{div}(f_P) = n[P] - n[\mathcal{O}] \quad \text{and} \quad \mathrm{div}(f_Q) = n[Q] - n[\mathcal{O}].$$

So, we require f_P and f_Q to have only P as a zero and only $\mathcal{O}$ as a pole, both of multiplicity n.
The Weil pairing is a function $e_n \colon E[n] \times E[n] \to \mu_n$, where μ_n denotes the nth roots of unity in $\mathbb{F}_q^*$, given by

$$e_n(P, Q) = \frac{f_P(Q + S)}{f_P(S)} \cdot \frac{f_Q(-S)}{f_Q(P - S)},$$

where S is any point on E such that $S \notin \{\mathcal{O}, P, -Q, P - Q\}$.

Now, consider an elliptic curve E over a prime field $\mathbb{F}_p$ and suppose $P, Q \in E(\mathbb{F}_p)$ and P has order ℓ. Let k be the smallest positive integer such that $\ell \mid p^k - 1$. In order to solve the ECDLP for points P and Q, we use the Weil pairing to map to points in $E(\mathbb{F}_p)$ onto elements of the field $\mathbb{F}_{p^k}$. To do this, we utilise the MOV algorithm which assumes we can solve the discrete logarithm problem in $\mathbb{F}_{p^k}$.

Protocol 3.14 (The MOV algorithm)**.** [13, p. 348]

(1) Compute the number of points on $E(\mathbb{F}_{p^k})$, we denote this by N.

(2) Choose a random $T \in E(\mathbb{F}_{p^k})$ satisfying $T \notin E(\mathbb{F}_p)$.

(3) Compute $T' = \frac{N}{\ell}T$. If $T' = \mathcal{O}$, repeat from (2). Otherwise, continue with T', a point of order ℓ.

(4) For chosen $P, Q \in E(\mathbb{F}_p)$, compute the Weil pairings

$$\alpha = e_\ell(P, T') \quad \text{and} \quad \beta = e_\ell(Q, T').$$

These are both in $\mathbb{F}_{p^k}^*$.

(5) Solve the discrete logarithm problem for α and β. That is, find an integer n such that $\beta = \alpha^n$.

(6) Then $Q = nP$ also, so the elliptic curve discrete logarithm problem has been solved for points P and Q.

Protocol 3.14 allows an intruder to easily solve the ECDLP using only the number of points on the elliptic curve over $\mathbb{F}_{p^k}$. The specific curve involved in any cryptosystem is a domain parameter, meaning it is almost always public information. Therefore it is vital to consider the value of k when deciding an elliptic curve since, if chosen poorly, it could allow an outside party to gain access to private information.

To prevent attacks using Weil pairings, an elliptic curve and finite field must be chosen such that the number of points on the curve is a large prime N, we usually use $N > 2^{160}$. This N must not divide $p^k - 1$ for every $k \in \{1, \ldots, C\}$ for some C large enough, $C = 20$ is common. This ensures we are unable to solve the DLP in $\mathbb{F}_{p^k}$, which is essential to the success of the MOV algorithm [8, p. 618].

A curve that is designed to resist any known attacks is Curve25519, introduced by Daniel J. Bernstein in 2006 [1, p. 210]. It is defined by the equation

$$y^2 = x^3 + 486662x^2 + x$$

over the prime field $\mathbb{F}_{2^{255}-19}$. It provides highly efficient and fast encryption, with any public and private keys being very short at only 32 bytes.

Curve25519's design minimises the risk from attacks like the MOV algorithm, making it particularly suited for high-security protocols where we wish to guarantee safe communication. A widely known theorem to find the number of points on an elliptic curve is Hasse's theorem.

Theorem 3.15 (Hasse)**.** [11, p. 138] *Let E be an elliptic curve defined over a finite field $\mathbb{F}_q$. Then*

$$|\#E(\mathbb{F}_q) - q - 1| \leq 2\sqrt{q}.$$

This theorem, proven by Hasse in the 1930s, provides a bound on the number of points on any elliptic curve over a prime field, allowing us to estimate the total within an error of $2\sqrt{q}$.

Example 3.16 (Using Hasse's theorem)**.** Consider the elliptic curve

$$E : y^2 = x^3 - 4x + 9$$

over the finite field $\mathbb{F}_{23}$. Let $N = \#E(\mathbb{F}_{23})$ be the number of points on the curve. Then, by Hasse's theorem, we have

$$|N - 24| \leq 2\sqrt{23}.$$

Since $2\sqrt{23} \approx 9.592$, we get $24 - 9.592 \leq N \leq 24 + 9.592$, which simplifies to $14.408 \leq N \leq 33.592$. Thus,

$$15 \leq N \leq 33.$$

So, the number of points on the curve must lie between 15 and 33.

3.6 Why ECC outperforms RSA cryptography

In contrast to elliptic curve cryptography, which relies on the ECDLP, Rivest-Shamir-Adleman (RSA) cryptography relies on the multiplication of two large prime numbers. The two original primes are then difficult to obtain from their product [2, pp. 150–151]. While this method is efficient, it is not as efficient as when using repeated point addition on an elliptic curve. These calculations are practically infeasible to reverse, making ECC an obvious choice for maximising security.

Elliptic curve cryptography has many advantages when compared to RSA cryptography. Firstly, the ECDLP is deemed more difficult to solve than any other problem, so an eavesdropper is less likely to successfully obtain private information. Elliptic curve cryptography also tends to require smaller key sizes than RSA cryptography, with a 256-bit ECC key providing almost the same strength as a 3072-bit RSA key [15, p. 5]. This reduces the amount of storage required to encrypt any message, making elliptic curve

cryptography ideal for modern systems like within mobile devices, where we require as much protection as possible in a compact device .

However, this increased security is accompanied by more complex computations, which may result in lower efficiency transmissions as the calculations involved in encrypting or decrypting a message are significantly more challenging. Nonetheless, when handling highly sensitive information it is valuable to dedicate more time to encryption if it will guarantee confidentiality.

While elliptic curve cryptography offers significant advantages, it is prone to future threats. The development of quantum computing, which involves applying quantum mechanics to cryptography, introduces many vulnerabilities. In particular, Shor's algorithm can solve the ECDLP in polynomial time, potentially making ECC completely ineffective [12, p. 483]. Therefore, research into post-quantum cryptographic protocols is vital to ensure secure encryption in the future.

A promising post-quantum alternative to elliptic curve cryptography is Supersingular Isogeny Key Exchange (SIKE), which uses the structure of elliptic curves but appears to resist attacks from quantum computers. Isogenies are mappings between elliptic curves which preserve their group structure. SIKE relies on the difficulty of finding isogenies between a specific type of curves, called supersingular elliptic curves [3, p. 24]. The development of hybrid cryptosystems like SIKE, which utilise the strengths of elliptic curve cryptography against many current attacks while also providing protection against emerging threats, is essential to post-quantum cryptography.

4 Conclusion

This essay has explored the various foundations behind elliptic curve cryptography, and how they can then be applied in a real system. Cryptography is essential in everyday life, from sending messages to saving files, security is key. Elliptic curve cryptography is an ever-growing subset of this which appears to provide more security using less storage. By gaining an insight into the elliptic curve group and the ECDLP, we have seen how elliptic curve cryptography offers many advantages when compared to other methods. We also introduced various applications of ECC, showcasing its versatility.

From use in cryptocurrencies like Bitcoin to protection of sensitive military information, elliptic curves truly are the most sustainable way forward as threats to security, like the Weil pairing, evolve. With the development of quantum cryptography, future research will be dedicated to optimising the efficiency of elliptic curve cryptography, hopefully providing an indestructible method of data encryption. Having only covered a few cryptosystems, it is essential to note that there are countless applications waiting to be discovered.

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All diagrams in this essay were produced using the Python programming language.